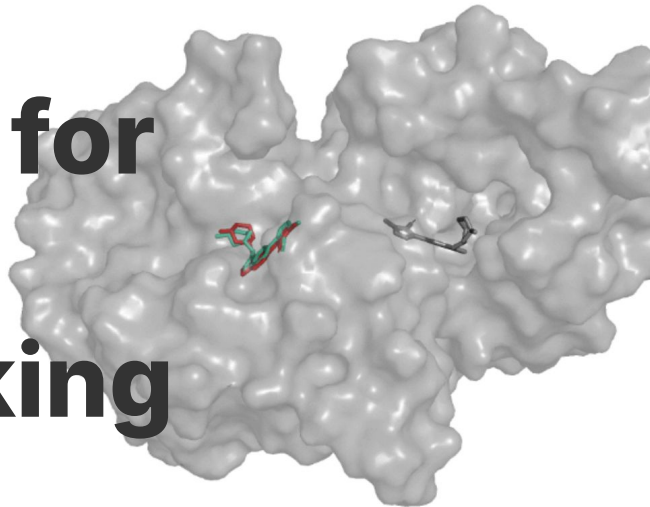


Geodesic Guidance for Diffusion-Based Protein-Ligand Docking



Incorporating Geometric Priors into Generative Processes on

$$\mathbb{T}(3) \times SO(3) \times SO(2)^m$$

Based on the paper "Informed protein-ligand docking via geodesic guidance in translational, rotational and torsional spaces." *Nature Machine Intelligence* (2025).

Score-Based Diffusion on $\mathbb{T}(3) \times SO(3) \times SO(2)^m$

Let $\mathbb{P} = \mathbb{T}(3) \times SO(3) \times SO(2)^m$ and $g = (r, R, \theta) \in \mathbb{P}$

Then, $T_g\mathbb{P} = \mathfrak{t}(3) \oplus \mathfrak{so}(3) \oplus \mathfrak{so}(2)^m \cong \mathbb{R}^3 \oplus \mathbb{R}^3 \oplus \mathbb{R}^m$

Diffusion kernels:

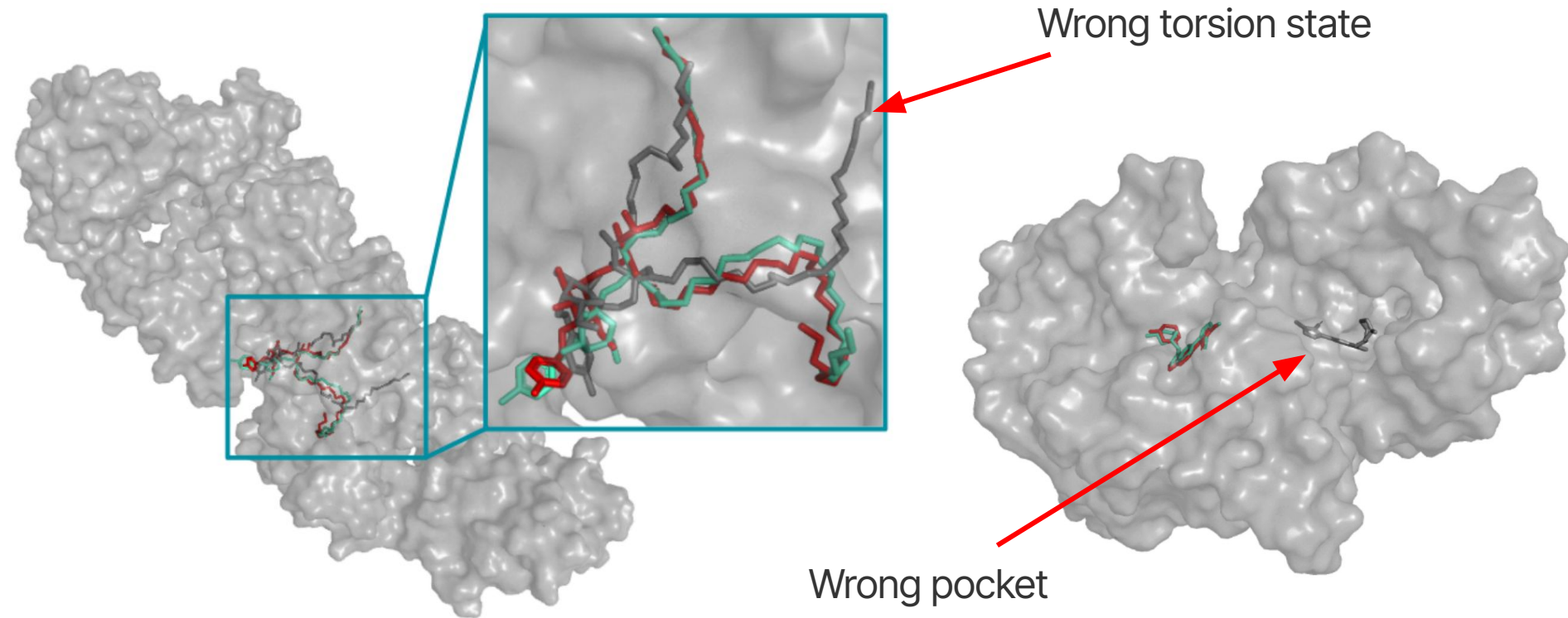
- Standard Gaussian on $\mathbb{T}(3)$
- Isotropic Gaussian on $SO(3)$
- Wrapped Gaussian on $SO(2)^m$

Loss function:

$$\mathcal{L} = \|s_\theta^{tr}(t, x) - \nabla \ln p_t^{tr}(\Delta r|0)\|^2 + \|s_\theta^{rot}(t, x) - \nabla \ln p_t^{rot}(\Delta R|0)\|^2 + \|s_\theta^{tor}(t, x) - \nabla \ln p_t^{tor}(\Delta \theta|0)\|^2$$

Navigating High-Entropy Manifolds

Grey: Diffdock
Blue: Ours
Red: Ground truth



Geodesic Guidance on the Product Manifold (I)

- Sequential composition of diffusion step and geodesic correction
- Consider maps $\exp : \mathfrak{p} \rightarrow \mathbb{P}$ $\ln : \mathbb{P} \rightarrow \mathfrak{p}$ (principal branch)

Diffdock $V_{\text{diffdock}}^k = g_k(t)^2 \cdot \Delta t \cdot s_{\theta}^k(x_t, t) + g_k(t) \cdot \sqrt{t} \cdot z$

where $V_{\text{diffdock}}^{\text{tr,rot}} \in \mathbb{R}^3$, $V_{\text{diffdock}}^{\text{tor}} \in \mathbb{R}^m$

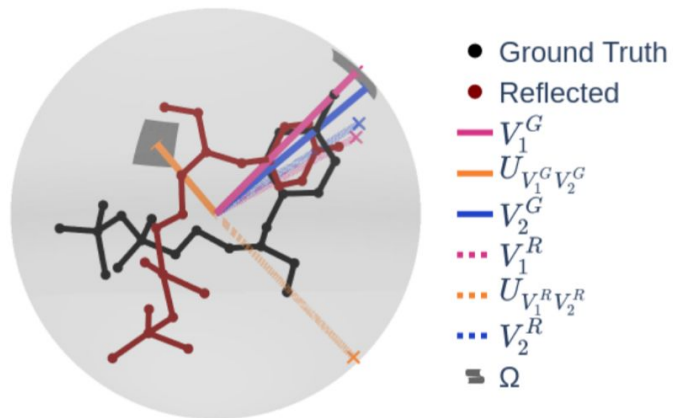
Geodesic guidance Consider prior region $\mathcal{R} \subseteq \mathbb{P}$, let $x_t \mapsto g_t \in \mathbb{P}$

$$g^* = \operatorname{argmin}_{g \in f(\mathcal{R})} \|\ln(gg_t^{-1})\|, \quad V_{\text{guidance}} = \ln(g^*g_t^{-1})$$

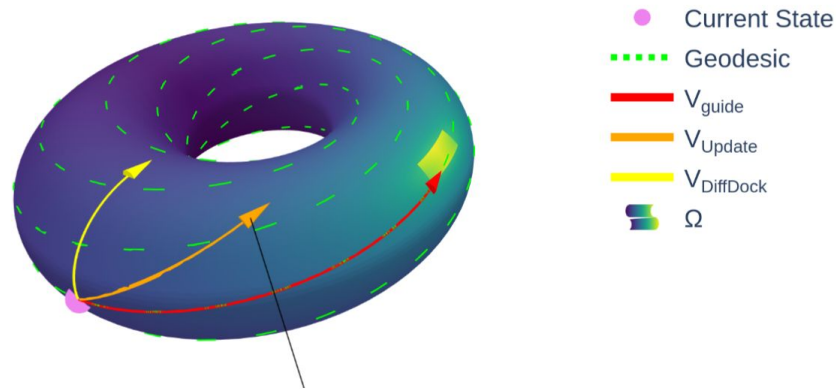
Update

$$g_{t-\Delta t} = \begin{cases} \exp(\gamma V_{\text{guidance}}) \exp((1 - \gamma)V_{\text{diffdock}})g_t & \text{if } g_t \notin \mathcal{R} \\ \exp(V_{\text{diffdock}})g_t & \text{if } g_t \in \mathcal{R} \end{cases} \mapsto x_{t-\Delta t}$$

Geodesic Guidance on the Product Manifold (II)



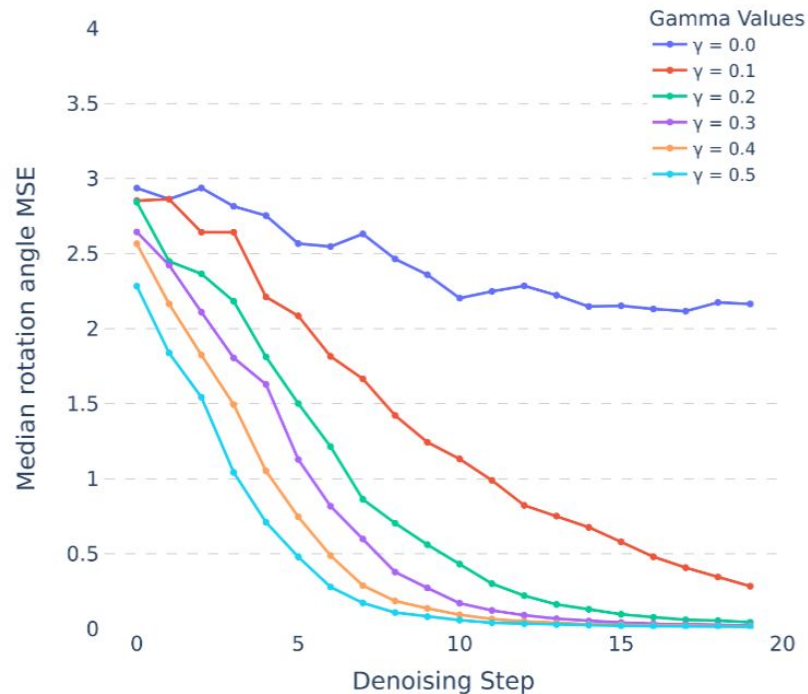
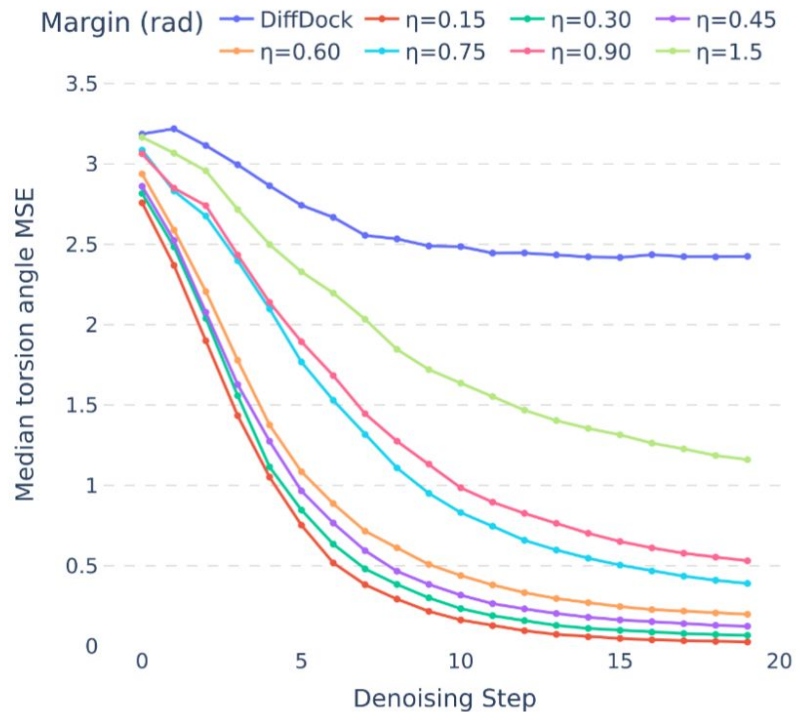
- Disambiguate rotational state
- Apply sequential rotations



$$\begin{aligned}\theta_{t-\Delta t} &= \theta_t + ((1 - \gamma)V_{\text{diffdock}} + \gamma V_{\text{guidance}}) \bmod 2\pi \\ &= ((\theta_t + (1 - \gamma)V_{\text{diffdock}}) \bmod 2\pi + \gamma V_{\text{guidance}}) \bmod 2\pi\end{aligned}$$

- Sequential update $\equiv$ convex combination on the Lie Algebra for $\mathbb{T}(3)$ and $SO(2)^m$

Exploration of Guidance Schemes and Fuzzy Priors



Concluding Remarks: the Resulting Distribution

- Controlled inductive bias
- Distributional concentration
- Inference time modularity

